

BigTreeASignalAiBot — Full Feature List

Core Features (Required)

1. RAG Q&A

- Files: bot/rag.py, bot/listener.py, bot/confidence.py
- Description: Users ask questions in Discord channels; the bot retrieves relevant content from the ChromaDB knowledge base and uses GPT to generate answers that imitate the channel owner's style
- Config: OPENAI_API_KEY, LLM_MODEL, RAG_TOP_K, RAG_MAX_DISTANCE, CONFIDENCE_THRESHOLD
- Usage: Automatic — users just post in a target channel

2. Confidence Routing

- Files: bot/confidence.py
- Description: Based on the confidence score, either auto-reply ($\geq$ threshold) or forward to the Owner for review ($<$ threshold)
- Config: CONFIDENCE_THRESHOLD=7 (default), RESPOND_MODE=auto|review|questions|mention_only

3. Owner Review System

- Files: bot/review.py, bot/review_queue.py
- Description: Low-confidence answers are sent to the Owner via DM with Approve / Edit / Reject buttons
- Usage: Owner clicks buttons in DM; Approved content is automatically learned into the knowledge base

4. Knowledge Base Ingestion

- Files: ingestion/ingest.py, ingestion/preprocess.py
- Description: Preprocess, chunk, and vectorize Discord-exported JSON data into ChromaDB
- Usage: python -m ingestion.ingest

5. Channel Listening & Message Handling

- Files: bot/listener.py
- Description: Listen to target channel messages, filter spam/politeness phrases, detect questions, and maintain conversation memory
- Config: TARGET_CHANNEL_IDS, EXCLUDED_CHANNEL_IDS, CONVERSATION_MEMORY_SIZE, CONVERSATION_MEMORY_TTL

Important Features (Strongly Recommended)

6. Rate Limiting

- Files: bot/listener.py
- Description: Per-user cooldown + global per-minute cap to prevent abuse
- Config: USER_COOLDOWN_SECONDS=30, GLOBAL_MAX_PER_MINUTE=10

7. Offline Backfill

- Files: bot/listener.py
- Description: After restart, automatically scan unanswered questions from the offline period
- Config: OFFLINE_BACKFILL_ENABLED=true, OFFLINE_BACKFILL_LOOKBACK_HOURS=24

8. Auto-Learn Owner Messages

- Files: bot/listener.py
- Description: Owner posts in channels are automatically ingested into ChromaDB so the knowledge base keeps growing
- Usage: Automatic — Owner posts normally

9. Thread Reply Support

- Files: bot/listener.py
- Description: Can reply inside Threads and automatically pull Thread context
- Config: `THREAD_AUTO_REPLY=true`, `THREAD_CONTEXT_MESSAGES=15`

10. Image / Chart Analysis

- Files: bot/rag.py, bot/listener.py
- Description: When users send images (e.g. candlestick charts), analyze them with GPT-4o Vision
- Config: `VISION_MODEL=gpt-4o`

11. Voice Message Transcription & Reply

- Files: bot/listener.py
- Description:
 - Owner voice: Discord voice clips are transcribed with Whisper and auto-learned into the knowledge base (no public reply)
 - Member voice: Only Discord voice clips are processed (not arbitrary mp3/songs). After Whisper transcription, RAG replies; the channel shows typing while transcribing
 - Limits: Over ~2MB or 90 seconds prompts the user to use text; after transcription, chit-chat like “thanks/ok” or non-questions are silently skipped without a full reply
 - Usage: Send a Discord voice message in a target channel; voice + text caption also works
 - Note: Bot must be able to read attachments; transcription depends on OpenAI Whisper

12. Stats Tracking

- Files: bot/stats.py
- Description: Record total queries, auto-replies, forwards, confidence, latency, etc., with time-range filters
- Usage: `/stats` command or Admin panel

13. Embedding Cache

- Files: bot/cache.py
- Description: LRU + TTL cache for embedding results to reduce duplicate API calls
- Usage: Enabled automatically

Optional Features (Enable as Needed)

14. Admin Web Panel

- Files: bot/admin.py
- Description: Browser-based admin panel for stats / config / knowledge base / FAQ
- Config: `ADMIN_ENABLED=true`, `ADMIN_PORT=8082`, `ADMIN_SECRET`
- Usage: Open `http://host:8082/admin/` in a browser

15. FastAPI Management API

- Files: bot/api/ directory
- Description: RESTful API + WebSocket for stats / config / review / knowledge base / FAQ / promo management
- Config: API_ENABLED=true, API_PORT=8090, API_USERNAME, API_PASSWORD
- Usage: Admin mobile app (app/) manages the bot through this API

16. Public Client API

- Files: bot/api/routes_public.py
- Description: Public API for end users (chat, image analysis, FAQ, search, summaries, etc.)
- Config: CLIENT_API_ENABLED=true, CLIENT_API_KEY, CLIENT_RATE_LIMIT_PER_MINUTE=20
- Usage: Web client (web-client/) and mobile client (app-client/) interact via this API

17. Web Client

- Files: web-client/ directory
- Description: React Web App — chat, search, summaries, FAQ, bookmarks, history
- Usage: npm run dev (dev) or npm run build (production)

18. Mobile Client

- Files: app-client/ directory
- Description: Expo React Native App — same features as the Web client
- Usage: Publish to App Store / Google Play via EAS Build

19. Admin Mobile App

- Files: app/ directory
- Description: Expo App for Owner remote management (review, stats, config, knowledge base)
- Usage: Connects through the FastAPI management API

20. Daily Digest

- Files: bot/digest.py
- Description: Daily scheduled 24-hour activity digest posted to a designated channel + DM to Owner
- Config: DIGEST_ENABLED=true, DIGEST_HOUR=22 (UTC), DIGEST_CHANNEL_ID

21. FAQ Auto-Generation

- Files: bot/faq.py
- Description: Cluster high-confidence queries with GPT to auto-generate FAQ entries
- Usage: /generate_faq command or Generate FAQ button in Admin panel

22. Promo Scheduling System

- Files: bot/scheduler.py, bot/commands.py, bot/promo_config.py
- Description: Schedule promo posts / signal recaps / educational content with repeat options (hourly/daily/weekly/monthly)
- Config: PROMO_ENABLED=true, PROMO_CHANNEL_IDS, SIGNAL_PRODUCT_NAME, SIGNAL_PRODUCT_URL
- Usage: /schedule_promo (optional dm_role to also DM the opt-in notify role), /schedule_trial, /schedule_lesson

23. User Testimonial Collection

- Files: bot/testimonials.py
- Description: Auto-detect positive user messages, forward to Owner for Approve/Reject, and display approved

testimonials

- Config: TESTIMONIAL_DETECTION_ENABLED=true, TESTIMONIAL_CHANNEL_ID
- Usage: /testimonials to view

24. Webhook Ingestion

- Files: bot/webhook.py
- Description: External systems ingest text into the knowledge base via HTTP POST
- Config: WEBHOOK_ENABLED=true, WEBHOOK_PORT=8081, WEBHOOK_SECRET

25. YouTube Ingestion

- Files: ingestion/ingest_youtube.py
- Description: Ingest YouTube video captions into ChromaDB (captions API first; if none, Whisper fallback: yt-dlp download audio → ffmpeg transcode → OpenAI Whisper)
- Usage: python -m ingestion.ingest_youtube --urls "URL"
- Dependencies: pip install 'yt-dlp[default]' imageio-ffmpeg; if system ffmpeg is missing, place ffmpeg/ffprobe (and deno required by newer yt-dlp) into the venv Scripts/ or bin/ directory. Code resolves binaries by full path and does not rely on system PATH

26. PDF Ingestion

- Files: ingestion/ingest_pdf.py
- Description: Ingest PDF documents into the knowledge base
- Usage: python -m ingestion.ingest_pdf --files "file.pdf"

27. Style Analysis

- Files: ingestion/analyze_style.py
- Description: Analyze Owner writing style and generate a style guide for RAG
- Usage: python -m ingestion.analyze_style

28. Scheduled Ingestion

- Files: bot/ingestion_scheduler.py
- Description: Background scheduled runs of ingestion and style analysis
- Config: INGEST_INTERVAL_HOURS, STYLE_INTERVAL_HOURS

29. Multilingual Support

- Files: bot/config.py
- Description: UI strings in Chinese (zh) and English (en)
- Config: BOT_LANGUAGE=zh

30. Negative Feedback Learning

- Files: bot/review.py, bot/rag.py
- Description: Owner-Rejected answers are saved as negative samples and injected into later prompts to avoid repeating mistakes
- Usage: Automatic; stored in data/negative_samples.json

Slash Commands Summary

Public Commands (All Users)

Command	Description	Usage
/ask	Ask the AI assistant (RAG knowledge base)	/ask question:your question
/signal	View BigTreeSignal product intro (not limited to promo channels)	/signal
/invite	Generate a personal invite link	/invite
/testimonials	Show recent user testimonials	/testimonials
/faq	View FAQ	/faq
/promo_notify	Opt in or out of promo DM notifications	/promo_notify action:opt-in

Owner-Only Commands

Promo management:

Command	Description	Usage
/post_promo	Post a promo immediately (optional sync DM)	/post_promo title:title description:content dm_role:@promo-notify
/schedule_promo	Schedule a promo (supports repeat; optional sync DM)	/schedule_promo title:title description:desc time:time repeat:mode dm_role:@promo-notify
/dm_role	Send promo DMs to the opt-in notify role	/dm_role role:@promo-notify title:title description:content
/promo_notify_panel	Post a promo DM subscription panel in a channel	/promo_notify_panel
/list_promos	List all scheduled promos	/list_promos
/cancel_promo	Cancel a scheduled promo	/cancel_promo promo_id:ID
/schedule_trial	Schedule a free signal recap post (supports repeat)	/schedule_trial title:title content:content time:time repeat:mode

Education management:

Command	Description	Usage
/schedule_lesson	Schedule an educational push (supports repeat)	/schedule_lesson title:title content:content time:time repeat:mode
/list_lessons	List all scheduled lessons	/list_lessons
/cancel_lesson	Cancel a scheduled lesson	/cancel_lesson lesson_id:ID
/resend_summary	Resend a YouTube video GPT summary (auto-ingest if transcript missing)	/resend_summary or /resend_summary video_url:link

Dev / Ops:

Command	Description	Usage
/status	Bot status (uptime, queue, knowledge base count)	/status
/stats	Query stats (totals, confidence, popular questions)	/stats
/search_kb	Search knowledge base documents	/search_kb query:keywords top_k:count
/generate_faq	Auto-generate FAQ from frequent questions	/generate_faq
/weekly_summary	Immediately generate and post this week's summary	/weekly_summary
/daily_summary	Immediately generate and post today's summary	/daily_summary

/views	[Owner] Scan owner posts across the server, briefly summarize, and post to this channel	/views hours:24
/funnel	View acquisition conversion funnel	/funnel days:7

Repeat Mode Options

Schedule commands (/schedule_promo, /schedule_trial, /schedule_lesson) all support these repeat modes:

Option	Description
No repeat	Send once and stop (default)
Hourly	Automatically resend every hour
Daily	Automatically resend every day
Weekly	Automatically resend every week
Monthly	Automatically resend every 30 days

Newer Features

31. Multilingual Auto-Detect & Reply

- Files: bot/lang_detect.py, bot/listener.py
- Description: Auto-detect user message language (zh/en/ja/ko) and instruct the LLM to reply in the same language
- Config: AUTO_LANG_DETECT=true (on by default)
- Usage: Users ask in any supported language; the bot answers in the same language

32. Knowledge Base Quality Report & /kb_report

- Files: bot/commands.py, bot/feedback.py, bot/leaderboard.py
- Description: Owner can view KB quality report: total docs, 30-day queries, low-confidence ratio, satisfaction, frequent questions
- Command: /kb_report

33. User Satisfaction Feedback ([up]/[down] Reactions)

- Files: bot/feedback.py, bot/listener.py
- Description: When users add thumbs-up/down reactions to bot replies, feedback is recorded; Owner can view stats via /satisfaction
- Config: FEEDBACK_ENABLED=true (on by default)
- Command: /satisfaction days:30

34. Conversation Summary & /pin_summary

- Files: bot/commands.py
- Description: Owner uses /pin_summary to summarize recent channel discussion, generate a summary, and pin it
- Command: /pin_summary count:20

35. Scheduled Reminders /schedule_reminder

- Files: bot/reminders.py, bot/scheduler.py, bot/commands.py
- Description: Owner can schedule custom reminder messages (e.g. market open, data releases) with repeat support

- Command: `/schedule_reminder`, `/list_reminders`, `/cancel_reminder`

36. VIP Role Recognition

- Files: `bot/listener.py`, `bot/config.py`
- Description: Configure VIP role IDs; users with those roles are exempt from rate limits
- Config: `VIP_ROLE_IDS=roleid1,roleid2`

37. Keyword Monitoring & Alerts

- Files: `bot/keyword_alert.py`, `bot/listener.py`, `bot/commands.py`
- Description: Owner configures watch keywords; when a user message contains one, the bot immediately DMs the Owner
- Config: `KEYWORD_ALERT_ENABLED=true` (on by default)
- Command: `/add_alert`, `/remove_alert`, `/list_alerts`

38. Knowledge Base Versioning & Rollback

- Files: `bot/kb_versioning.py`, `bot/commands.py`
- Description: KB snapshot management; snapshots can be created before ingest; Owner can browse historical snapshots
- Command: `/kb_snapshots`

39. Enhanced Welcome Flow

- Files: `bot/welcome_flow.py`, `bot/listener.py`
- Description: Multi-step welcome DM for new members: immediate welcome (with `/faq` `/ask` `/signal`). If `PROMO_NOTIFY_ROLE_IDS` is configured, the same welcome DM includes “Opt in / Unsubscribe” buttons; members only join the promo DM list after clicking
- Config: `WELCOME_FLOW_ENABLED=true` (recommended), `WELCOME_STEP2_DELAY` replaced by drip delays: `WELCOME_VALUE_DELAY_SECONDS=14400` (value content after 4 hours), `WELCOME_CTA_DELAY_SECONDS=86400` (product CTA next day), `WELCOME_REMINDER_DELAY_SECONDS=259200` (day-3 reminder)

40. Leaderboard Command

- Files: `bot/leaderboard.py`, `bot/commands.py`
- Description: View most active channels, confidence distribution, and other ranked stats
- Command: `/leaderboard days:30`

41. A/B Test Reply Styles

- Files: `bot/ab_test.py`, `bot/commands.py`
- Description: When enabled, each reply randomly picks a style variant and tracks satisfaction per variant
- Config: `AB_TEST_ENABLED=false` (off by default), `AB_TEST_VARIANTS=casual:style_a.txt,formal:style_b.txt`
- Command: `/ab_results`

42. Export Conversations

- Files: `bot/export.py`, `bot/commands.py`
- Description: Owner can export conversation logs as JSON or CSV
- Command: `/export_conversations format:JSON days:30`

Newer Slash Commands Quick Reference

Command	Description	Usage
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/schedule_reminder	Schedule a reminder message	/schedule_reminder title:title message:content time:2025-01-01 09:00
/list_reminders	List all reminders	/list_reminders
/cancel_reminder	Cancel a reminder	/cancel_reminder reminder_id:ID
/add_alert	Add keyword alert	/add_alert keyword:keyword
/remove_alert	Remove keyword alert	/remove_alert keyword:keyword
/list_alerts	List monitored keywords	/list_alerts
/kb_report	Knowledge base quality report	/kb_report
/kb_snapshots	View knowledge base snapshots	/kb_snapshots
/leaderboard	Activity leaderboard	/leaderboard days:30
/ab_results	A/B test results	/ab_results
/export_conversations	Export conversations	/export_conversations format:JSON days:30
/pin_summary	Channel discussion summary	/pin_summary count:20
/satisfaction	Satisfaction stats	/satisfaction days:30
/weekly_summary	Immediately generate and post this week's summary	/weekly_summary
/daily_summary	Immediately generate and post today's summary	/daily_summary
/views	[Owner] Scan owner posts across the server, briefly summarize, and post here	/views hours:24
/funnel	Acquisition conversion funnel	/funnel days:7
/invite	Generate a personal invite link	/invite
/promo_notify	Opt in or out of promo DM notifications	/promo_notify action:opt-in
/promo_notify_panel	[Owner] Post promo DM subscription panel	/promo_notify_panel
/dm_role	[Owner] Send promo DMs to opt-in notify role	/dm_role role:@promo-notify title:title description:content
/resend_summary	[Owner] Resend YouTube video summary (auto lookup/ingest)	/resend_summary or video_url:https://youtu.be/xxx

Environment Variables Quick Reference

All settings are configured via the .env file; see .env.example for a full sample.

Advanced Enhancements

43. Clarification Follow-up After Auto-Reply

- Files: bot/clarification.py, bot/listener.py, bot/config.py
- Description: When confidence is lower but still eligible for auto-reply, ask 1 clarifying question first (e.g. timeframe / risk preference) instead of outputting a vague conclusion
- Config: FEATURE_CLARIFICATION_FOLLOWUP, CLARIFICATION_CONFIDENCE_MAX, FEATURE_CLARIFICATION_CANARY_CHANNEL_IDS

44. Session Memory Summary

- Files: bot/session_summary.py, bot/listener.py, bot/config.py
- Description: In long conversations, automatically compress older messages into a short summary while keeping the most recent turns verbatim, improving context stability and reducing token use
- Config: FEATURE_SESSION_SUMMARY, SESSION_SUMMARY_TRIGGER_MESSAGES, SESSION_SUMMARY_KEEP_RECENT,

FEATURE_SESSION_SUMMARY_CANARY_CHANNEL_IDS

45. Feature Flags + Canary Channels

- Files: bot/feature_flags.py, bot/config.py
- Description: Per-channel canary rollout. If the canary list is empty, the feature is enabled everywhere; otherwise only in canary channels
- Example: FEATURE_LANG_DETECT=true, FEATURE_LANG_DETECT_CANARY_CHANNEL_IDS=123,456

46. High-Risk Answer Safety Guardrails

- Files: bot/guardrails.py, bot/listener.py, bot/config.py
- Description: Detect high-risk phrasing (e.g. all-in, guaranteed returns, max position / full portfolio) and either force owner review or append a risk disclaimer template
- Config: FEATURE_SAFETY_GUARDRAILS, GUARDRAIL_MODE=force_review|disclaimer, GUARDRAIL_DISCLAIMER, FEATURE_SAFETY_GUARDRAILS_CANARY_CHANNEL_IDS

47. Active Learning Loop (Feedback-to-Ingestion Pipeline)

- Files: bot/feedback_learning.py, bot/listener.py, bot/review.py, bot/scheduler.py
- Description: Automatically collect [down] feedback and owner-edited replies into a “KB gaps queue,” and daily summarize Top 10 missing questions
- Config: FEATURE_FEEDBACK_LEARNING, FEATURE_FEEDBACK_LEARNING_CANARY_CHANNEL_IDS

48. SLA / Reliability Monitoring

- Files: bot/reliability.py, bot/rag.py, bot/scheduler.py, bot/config.py
- Description: Monitor p95 latency, OpenAI error rate, review queue backlog, scheduler tick lag; alert the owner when thresholds are exceeded
- Config: FEATURE_SLA_MONITORING, SLA_P95_LATENCY_MS_THRESHOLD, SLA_OPENAI_ERROR_RATE_THRESHOLD, SLA_REVIEW_QUEUE_BACKLOG_THRESHOLD, SLA_SCHEDULER_MISS_SECONDS, SLA_ALERT_COOLDOWN_SECONDS

49. Jin10 Market Flash News Feed

- Files: bot/news_feed.py, bot/config.py
- Description: Poll Jin10 Flash API every 30 seconds and push important market flashes (e.g. central bank decisions, NFP, breaking events) to designated Discord channels. Important flashes use Embed rich format; normal flashes are plain text. Supports dedupe, ad filtering, multi-channel push. On bot reconnect, backfills important flashes missed while offline (default last 24 hours, max 50). last_id advances only after a successful post to avoid gaps
- Config: NEWS_FEED_ENABLED=true, NEWS_CHANNEL_IDS=channelId1,channelId2, NEWS_POLL_INTERVAL_SECONDS=30, NEWS_IMPORTANT_ONLY=true, NEWS_BACKFILL_HOURS=24 (if offline longer than 24 hours or more than 50 important flashes, older ones are not backfilled)

50. Weekly Summary

- Files: bot/weekly_summary.py, bot/config.py
- Description: On a configurable weekday at a configured ET time, collect all owner messages and owner replies from designated channels for the week, generate a key summary with GPT, post as Embed to designated channels and @everyone. Reply messages automatically pull the original question as context. Supports multi-channel scan and multi-channel publish. Reliability: channel read / network failures are no longer misclassified as “no messages this week” ; retries every 30 minutes after failure; still runs within 18 hours if the scheduled time was missed; sleep uses chunked timing to reduce overslept schedules after PC sleep
- Config: WEEKLY_SUMMARY_ENABLED=true, WEEKLY_SUMMARY_CHANNELS=channelId1,channelId2 (scan sources), WEEKLY_SUMMARY_DAY=5 (0=Mon, 5=Sat), WEEKLY_SUMMARY_HOUR=20 (ET), WEEKLY_SUMMARY_MINUTE=5, WEEKLY_SUMMARY_POST_CHANNELS=channelId (publish targets; empty falls back to SUMMARY_CHANNELS)

- Command: `/weekly_summary` (Owner manual trigger; can immediately resend last week's missed summary)

51. Promo Auto-Scheduling (Promo Monitor)

- Files: `bot/promo_monitor.py`, `bot/scheduler.py`, `bot/config.py`
- Description: When Owner posts promo info in a designated source channel, the bot automatically cancels old promo schedules, creates a new daily-repeat `schedule_promo` with `@everyone`, pushes to target channels at a configured time (default 4 PM ET), valid for 3 months (configurable). Auto-cancels on expiry. Image attachments are automatically embedded
- Config: `PROMO_MONITOR_ENABLED=true`, `PROMO_SOURCE_CHANNEL_ID=channelId`, `PROMO_PUSH_HOUR=16`, `PROMO_DURATION_DAYS=90`, `PROMO_PUSH_CHANNELS=channelId1, channelId2`

52. YouTube New Video Auto-Push (YouTube Monitor)

- Files: `bot/youtube_monitor.py`, `bot/config.py`
- Description: Daily at a configured time, poll the YouTube channel RSS feed; on new video, cancel old YouTube lesson schedules and create a new daily-repeat `schedule_lesson` pushed to Discord channels at a configured time (default 4 PM ET). Push includes title and watch link. If `auto-ingest` is on, captions/Whisper transcript go into the KB and a GPT summary is pushed to `YOUTUBE_SUMMARY_CHANNELS`. If transcription fails, a "new video notice" Embed is still sent; already-recorded videos need `/resend_summary` to resend the summary (see item 56)
- Config: `YOUTUBE_MONITOR_ENABLED=true`, `YOUTUBE_CHANNEL_ID=UCxxxxxxxxxx`, `YOUTUBE_CHECK_HOUR=11`, `YOUTUBE_CHECK_MINUTE=05`, `YOUTUBE_LESSON_PUSH_HOUR=16`, `YOUTUBE_LESSON_PUSH_CHANNELS=channelId1, channelId2`, `YOUTUBE_AUTO_INGEST=true`, `YOUTUBE_SUMMARY_CHANNELS=channelId`

53. Daily Summary

- Files: `bot/daily_summary.py`, `bot/config.py`
- Description: On each workday (Mon-Fri, configurable) at a configured time, collect that day's owner messages and replies, generate today's key summary with GPT, post as Embed to designated channels and `@everyone`. Fully independent from Weekly Summary. Posts once per day: if the timer fires early (e.g. 15:59) and already sent, it will not send again in the same minute
- Config: `DAILY_SUMMARY_ENABLED=true`, `DAILY_SUMMARY_CHANNELS=channelId1, channelId2` (scan sources), `DAILY_SUMMARY_DAYS=0, 1, 2, 3, 4` (0=Mon..6=Sun, default Mon-Fri), `DAILY_SUMMARY_HOUR=16` (ET), `DAILY_SUMMARY_MINUTE=0`, `DAILY_SUMMARY_POST_CHANNELS=channelId` (publish targets; empty falls back to `DAILY_SUMMARY_CHANNELS`)
- Command: `/daily_summary` (Owner manual trigger)

54. Auto Moderation

- Files: `bot/auto_mod.py`, `bot/ban_words.py`, `bot/config.py`
- Description: Automatically detect and delete spam, scam, and ads. Supports 7 detection layers:
 1. Keyword matching — zh/en spam/scam/ad phrases (free signals, guaranteed profit, etc.)
 2. Ban-word list (exact match) — Owner can dynamically add ban words via `/add_ban_word`; messages containing them are deleted
 3. Ban-word list (semantic match) — OpenAI embeddings for semantic similarity; different wording of the same meaning is still blocked (threshold configurable, default 0.82)
 4. External platform links — Telegram, WhatsApp, Line, short links, etc.
 5. Discord invite links — discord.gg links from non-exempt users
 6. Mass @ mentions / link spam — blocked above threshold
 7. Duplicate spam — same user repeating the same content in a short window
- Ban-word management commands:
 - `/add_ban_word <word>` — add a ban word (auto-computes embedding for semantic match)
 - `/remove_ban_word <word>` — remove a ban word
 - `/list_ban_words` — list all ban words
- Exemptions: Owner (`OWNER_USER_ID`), Bot, and designated roles (`AUTO_MOD_EXEMPT_ROLE_IDS`) are exempt

- Logging: Deletes are logged as Embeds to a designated log channel (author, channel, reason, content preview), and Owner is DMed (deleted content + sender Discord ID)
- Config: `AUTO_MOD_ENABLED=true`, `AUTO_MOD_LOG_CHANNEL_ID=channelId`, `AUTO_MOD_EXEMPT_ROLE_IDS=roleId1,roleId2`, `AUTO_MOD_MAX_MENTIONS=8`, `AUTO_MOD_MAX_LINKS=5`, `AUTO_MOD_DUP_WINDOW=60` (seconds), `AUTO_MOD_DUP_THRESHOLD=3`, `AUTO_MOD_BAN_WORDS_FILE=data/ban_words.json`, `AUTO_MOD_BAN_WORDS_SIMILARITY=0.82`
- Permission required: Bot needs Manage Messages
- Persistence: Ban-word list stored in `data/ban_words.json` with words and embeddings; loaded automatically on restart

55. Channel Topic Guard

- Files: `bot/topic_guard.py`, `bot/auto_mod.py`, `bot/config.py`
- Description: Enforce topic limits on designated channels — only allow investment-related discussion (trading, signals, stocks, FX, crypto, technical analysis, etc.). Chit-chat, nonsense, and offensive posts are auto-deleted
- Classification: GPT-4o-mini classifies each message as `on_topic` (allow), `off_topic` (delete), or `offensive` (delete); results cached 5 minutes to avoid duplicate API calls
- Pre-filter: Pure emoji or 1-2 character messages are treated as `off_topic` without calling GPT
- Fault tolerance: On GPT failure, default to allow (`on_topic`) to avoid false deletes
- Exemptions: Owner, Bot, and designated roles are exempt (inherits Auto Moderation exemptions)
- Config: `TOPIC_RESTRICTED_CHANNEL_IDS=channelId1,channelId2` (empty = disabled)
- Relation to Auto Mod: Topic Guard is the final Auto Moderation layer — after site-wide spam/ban-word filters, restricted channels get a topic check

56. Resend YouTube Summary

- Files: `bot/youtube_monitor.py`, `scripts/resend_youtube_summary.py`, `ingestion/ingest_youtube.py`
- Command: `/resend_summary`
- Description: Owner uses a slash command to resend a YouTube video GPT summary to `YOUTUBE_SUMMARY_CHANNELS`. Auto-checks ChromaDB: if transcript exists, summarize directly; if not, Whisper ingest then summarize; if still failing, generate a teaser summary from the title (gold Embed + footer note). Progress updates on the same ephemeral message; send failures are not reported as success. Only one resend job at a time. Works even when Monitor is off. `scripts/resend_youtube_summary.py` is a CLI backup (ingested videos only)
- Config: `YOUTUBE_SUMMARY_CHANNELS=channelId`, `OPENAI_API_KEY`, `LLM_MODEL`, `OWNER_USER_ID`
- Whisper dependencies (required when no captions): `pip install 'yt-dlp[default]' imageio-ffmpeg`; on Windows, place `ffmpeg.exe`, `ffprobe.exe`, `deno.exe` in `.venv/Scripts/` (newer yt-dlp needs a JS runtime to download YouTube)
- Usage (Slash command):

```
/resend_summary                — resend the most recently detected video
/resend_summary video_url:https://youtu.be/xxxxx — resend a specific video
/resend_summary video_url:... title:custom title  — override title
```

- Usage (script backup):

```
python scripts/resend_youtube_summary.py
python scripts/resend_youtube_summary.py --video-id nTWo8Wv7Jao
python scripts/resend_youtube_summary.py --video-id nTWo8Wv7Jao --dry-run
```

- Flow: ChromaDB has transcript → summarize directly; not ingested → auto Whisper ingest → summarize; transcription fails → teaser summary from title → post to summary channels (with acquisition CTA buttons; see item 59)

57. Purchase Intent Auto-Conversion

- Files: bot/acquisition.py, bot/listener.py
- Description: When users ask about subscription, price, trial, VIP, or how to buy, the bot skips normal RAG and immediately replies with a product Embed + CTA buttons (Learn product / Request trial / View FAQ), and DMs the Owner. Optionally assigns an “Inquiring” role for follow-up
- Config: INTENT_CONVERT_ENABLED=true, INTENT_NOTIFY_OWNER=true, INTENT_LEAD_ROLE_ID=roleId (0 = do not assign role), SIGNAL_PRODUCT_URL, FREE_TRIAL_ENABLED, FREE_TRIAL_URL
- Usage: Automatic. Placeholder links (e.g. your-product-url.com) will not generate buttons — replace with real URLs

58. New-Member 24 - 72 Hour Conversion Drip

- Files: bot/welcome_flow.py, bot/acquisition.py, bot/acquisition_cog.py
- Description: Immediate welcome DM after join (with /faq /ask /signal). If PROMO_NOTIFY_ROLE_IDS is set, welcome DM includes Opt in / Unsubscribe buttons. Then delayed sends: latest daily/video summary (prove value) → product CTA + testimonials → light day-3 reminder. Tasks persist in data/welcome_drip.json across restarts. VIP role holders stop further drip
- Config: WELCOME_FLOW_ENABLED=true, WELCOME_VALUE_DELAY_SECONDS=14400 (4 hours), WELCOME_CTA_DELAY_SECONDS=86400 (next day), WELCOME_REMINDER_DELAY_SECONDS=259200 (day 3)
- Usage: With WELCOME_FLOW_ENABLED on, runs for every new member. Delays are in seconds from join time

59. YouTube Summary Conversion Buttons

- Files: bot/youtube_monitor.py, bot/acquisition.py, scripts/resend_youtube_summary.py
- Description: Video summary Embeds include bottom buttons “Learn BigTreeSignal / Request trial / View FAQ,” so educational content also acquires leads. Auto-push and /resend_summary (plus the resend script) all include these buttons
- Config: SIGNAL_PRODUCT_URL, FREE_TRIAL_ENABLED, FREE_TRIAL_URL, YOUTUBE_SUMMARY_CHANNELS
- Usage: Sent with summaries after new-video auto-ingest; resend see item 56

60. Invite Virality / Personal Invite Links

- Files: bot/acquisition.py, bot/acquisition_cog.py, bot/commands.py
- Description: Members use /invite to create a personal Discord invite. On join, the bot compares invite use counts and records the inviter. Reaching a threshold can auto-grant a reward role. Requires create/view invite permissions and Server Members Intent
- Config: INVITE_TRACKING_ENABLED=true, INVITE_REWARD_THRESHOLD=3, INVITE_REWARD_ROLE_ID=roleId (0 = no reward role)
- Command: /invite
- Usage: Members run /invite in any server channel and get an ephemeral personal link

61. Conversion Funnel Dashboard

- Files: bot/acquisition.py, bot/commands.py
- Description: Track joins, welcome DM success/denied, /signal uses, purchase-intent hits, CTA sends, invite joins. Data in data/funnel.json
- Command: /funnel days:7 (Owner)
- Usage: Owner privately views last N days (1-90) funnel plus all-time totals. /signal is public and not limited to PROMO_CHANNEL_IDS

62. Opt-In Notify Role Promo DMs

- Files: bot/role_dm.py, bot/commands.py, bot/scheduler.py, bot/welcome_flow.py
- Description: Only send promo DMs to members who voluntarily opted into a notify role. Do not mass-DM manually tagged ops/interest roles (Discord treats unsolicited bulk DMs as spam). Roles outside the whitelist are rejected by the command.

- Config: `PROMO_NOTIFY_ROLE_IDS` (opt-in “promo notify” role IDs, comma-separated), `PROMO_DM_DELAY_SECONDS=1.2`, `PROMO_DM_MAX_RECIPIENTS=200`
- Usage:
 1. Create a “Promo Notify” role in Discord (no admin permissions), put its ID in `PROMO_NOTIFY_ROLE_IDS`; Bot role must be above it and have Manage Roles
 2. Owner runs `/promo_notify_panel` in an announcement channel (pin recommended); members click “Opt in” to join the list
 3. On join, welcome DM also includes the same Opt in / Unsubscribe buttons (requires `WELCOME_FLOW_ENABLED=true`)
 4. Send promos: `/dm_role` for DMs only (confirm recipient count first); or `/post_promo` / `/schedule_promo` with optional `dm_role` to sync DMs
 5. Unsubscribe only removes the notify role; existing ops tags are untouched. Sends exceeding `PROMO_DM_MAX_RECIPIENTS` are rejected
- Commands: `/promo_notify_panel`, `/promo_notify`, `/dm_role`; `dm_role` param on `/post_promo` and `/schedule_promo`
- Activation & subscription playbook: `GROWTH_PLAYBOOK.md`

63. Owner Views Summary (`/views`)

- Files: `bot/commands.py`, `bot/views_summary.py`, `bot/weekly_summary.py`, `bot/listener.py`
- Description: Owner runs `/views` in a server text channel or thread; the bot scans the whole server for your recent posts, uses GPT to generate a short, clear key-points summary (about ≤ 1200 characters, at most 3 subheadings), and publicly posts it in the current channel without @everyone. Collection scope:
 - All server text channels / forum posts / public threads: your posts, plus replies to members (reply targets anonymized as “member,” no nicknames)
 - Private threads / bot-visible group DMs you participated in: primarily your posts; counterparts keep only very short anonymous context — public summary does not name anyone or paste originals
 - Discord does not allow bots to read 1:1 DMs between you and members; to include 1:1 conversations, open a private server thread visible to the bot
 - Each channel pages newest→oldest until the time window ends, counting by owner message count so busy channels do not crowd out your posts
 - Excluded channels (`EXCLUDED_CHANNEL_IDS`) and promo source channel (`PROMO_SOURCE_CHANNEL_ID`) are not scanned
- Usage: `/views` (default last 24 hours); `/views hours:48` (1-168 hours). Owner only; server text channels/threads only
- Command: `/views hours:24`
- Note: Unlike `/daily_summary` and `/pin_summary`: summarizes only the channel owner's views and posts immediately in the command channel